

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$. Even though larger widths admit smaller λ at comparable construction error (Figure 2b), small slopes can delay readout optimization. We quantify this in the dictionary of Section 3.1, replacing the direct normalized least-squares solve with GD while keeping the features fixed.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$ and nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, μ_i are the descending readout-kernel eigenvalues, p_i is the fraction of target energy in eigenvector i , and $\bar{\rho}_i(\gamma) \in [0, 1]$ bounds μ_i/μ_1 from above. The explicit gamma-dependent rate bounds and finite-geometry allowances are given in Appendix A.

Proof sketch. Gamma enters the feature correlations through $d \coth(\gamma d)$, where d is the separation between training points. Controlling the finite-center corrections gives the relative-rate bounds; a standard spectral argument then yields the output-error inequality.

Increasing λ from $1/32$ to $1/4$ raises the thirtieth relative eigenvalue by 1.4×10^5 , versus only 1.3 for the fourth (Figure 3); the thirtieth mode carries about 2.5% of target energy at both endpoints. At $\lambda = 1/32$, a direct readout fit attains 6.2×10^{-10} dense-grid relative error, yet five million GD updates retain training error 0.210 against the theorem’s lower bound 0.2098 (Figure 4A). At $\lambda = 1/4$, GD reaches 3.88×10^{-4} over the same budget. Across the four bandwidths, the bound remains at least 75% of executed error throughout training: small slopes delay accuracy already available in the dictionary.

Over the same budget, joint Adam reaches median dense-grid relative error 1.81×10^{-3} , versus 6.62×10^{-7} on the supplied $\lambda = 1/4$ features, despite partial slope acquisition (Figure 4B–C). This motivates the next analysis: training must acquire useful slopes and centers together. The construction supplies both the feature geometry and the readout directly.

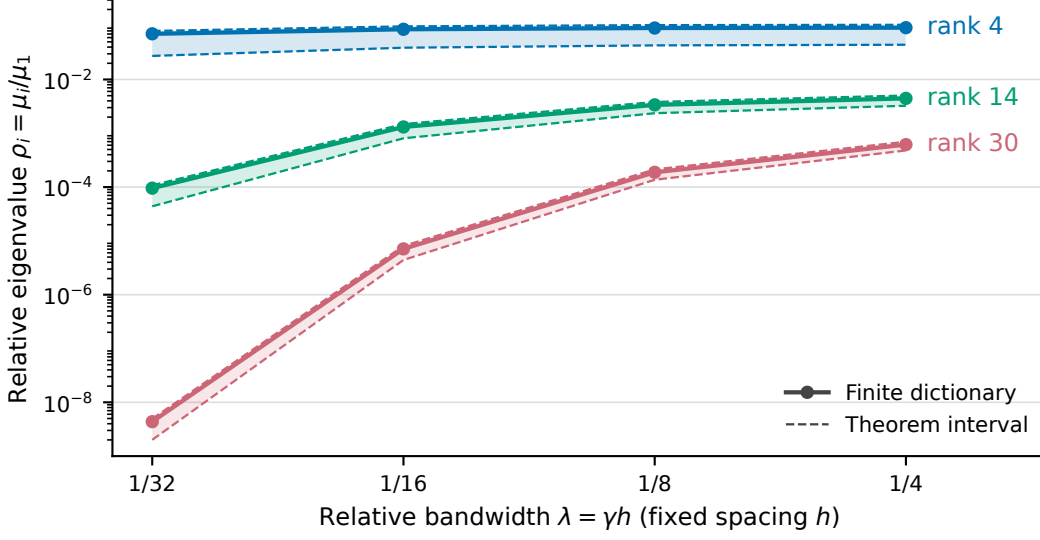


Figure 3: **Slope changes relative rates by very different amounts.** Finite-kernel ratios (solid, dots) and theorem intervals (dashed boundaries, shading) for three illustrative ordered ranks, at fixed width 512. Four bandwidths are evaluated; connecting lines are guides. Rank 30 carries 2.6% and 2.5% of target energy at the two endpoints, while its relative rate increases by 1.4×10^5 . Ordered rank does not identify the same eigenvector across slopes. All figures use the target and geometry in Appendix B.

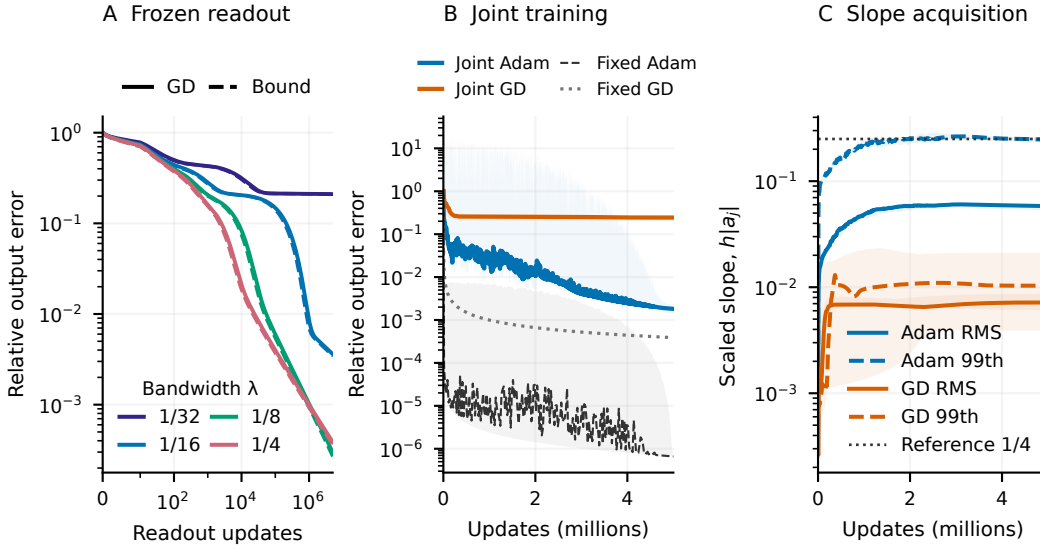


Figure 4: **From readout delay to feature acquisition.** **A:** executed frozen-readout GD errors (solid) and the square roots of the theorem’s lower bounds (dashed). **B:** joint Adam/GD errors, with frozen uniform $\lambda = 1/4$ references over the same five-million-update budget. **C:** RMS and 99th-percentile scaled slopes $h|a_j|$ from B’s joint runs. The reference bandwidth is a supplied geometry, not a necessary threshold for heterogeneous features. Joint curves summarize five paired seeds; shading retains seed variation and recorded within-bin extrema. The full horizon is shown and cosine schedules span that horizon. Appendix B specifies the error measure and selection protocol. The theorem applies to A; B–C are empirical.

A Explicit spectral bounds and proof

The main theorem concerns the actual finite tanh dictionary. An auxiliary center integral exposes its gamma dependence; lattice and exterior-center allowances transfer its spectrum to that dictionary. This appendix specifies each approximation and the resulting endpoints. The scale-dependent spectral bias is related to [Xu et al. \(2020\)](#); [Tancik et al. \(2020\)](#); the argument below quantifies it for the finite quasi-interpolant dictionary.

Finite setup. Let $z_1, \dots, z_M$ be distinct training points with $M \geq 2$, W consecutive centers on $c_0 + h\mathbb{Z}$ cover their interval, and $\gamma > 0$. Define

$$\Phi_\gamma = [\mathbf{1}, (\tanh(\gamma(z_a - x_j)))_{a,j}], \quad K_\gamma = \frac{1}{M} \Phi_\gamma \Phi_\gamma^\top, \quad y = (f(z_a))_{a=1}^M \neq 0. \quad (2)$$

Readout GD minimizes $\|\Phi_\gamma w - y\|^2 / (2M)$. Write its descending kernel eigenpairs as (μ_i, u_i) and define

$$\rho_i = \mu_i / \mu_1, \quad p_i = |u_i^\top y|^2 / \|y\|^2, \quad e_n^2 = \frac{\|\Phi_\gamma w_n - y\|^2}{\|y\|^2} = \frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2}. \quad (3)$$

The energy weights sum to one, including any exact nullspace component. The theorem permits arbitrary distinct samples; the reported experiments use uniform samples. Write $v_0 = \mathbf{1}/\sqrt{M}$ and let Q have orthonormal columns spanning $v_0^\perp$. This projection removes the common constant sample pattern, not any target assumption. Put $t_\gamma(c) = (\tanh(\gamma(z_a - c)))_a$.

The gamma-dependent correlation model. Its center-integrated, mean-zero kernel is

$$H_\gamma = \frac{1}{hM} \int_{\mathbb{R}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q \, dc = -\frac{2}{hM} Q^\top [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q, \quad d_{ab} = z_a - z_b. \quad (4)$$

The diagonal of the bracketed matrix is $1/\gamma$. Choose a finite set $\mathcal{F}$ of exterior lattice centers absent from the actual dictionary, leaving two unretained tails starting at $c_L < \min z_a$ and $c_R > \max z_a$. Subtract their retained contribution:

$$T_\gamma = \frac{1}{M} \sum_{c \in \mathcal{F}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q, \quad S_\gamma = H_\gamma - T_\gamma. \quad (5)$$

Let $\beta_1 \geq \dots \geq \beta_{M-1}$ be the eigenvalues of S_γ . For any $0 < \theta < \pi/2$, the lattice and remaining-tail allowances are

$$\delta_\gamma = \frac{8\gamma \|z - \bar{z}\mathbf{1}\|^2 \sec^4 \theta}{3hM [\exp(2\pi\theta/(\gamma h)) - 1]}, \quad \bar{z} = M^{-1} \sum_a z_a, \quad (6)$$

$$\tau_\gamma = \frac{4[e^{-4\gamma(c_R - z_{\max})} + e^{-4\gamma(z_{\min} - c_L)}]}{1 - e^{-4\gamma h}}. \quad (7)$$

They give the matrix enclosure

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^\top K_\gamma Q \preceq S_\gamma + \delta_\gamma I. \quad (8)$$

Relative-eigenvalue endpoints. Normalize using scalar bounds $\ell_\gamma \leq \mu_1 \leq L_\gamma$. The following require only matrix-vector products with the finite dictionary:

$$\ell_\gamma = v_0^\top K_\gamma v_0 \geq 1, \quad b_\gamma = \|Q^\top K_\gamma v_0\|, \quad c_\gamma = \beta_1 + \delta_\gamma, \quad (9)$$

$$L_\gamma = \frac{\ell_\gamma + c_\gamma + \sqrt{(\ell_\gamma - c_\gamma)^2 + 4b_\gamma^2}}{2}. \quad (10)$$

Set $\rho_1 = \bar{\rho}_1 = 1$ and $\beta_M = 0$. The relative-rate endpoints, including the upper bounds used in Theorem 3.2, are

$$\boxed{\rho_i = \frac{[\beta_i - \delta_\gamma - \tau_\gamma]_+}{L_\gamma}, \quad \bar{\rho}_i = \min\left\{1, \frac{\beta_{i-1} + \delta_\gamma}{\ell_\gamma}\right\}, \quad 2 \leq i \leq M.} \quad (11)$$

They enclose every finite-kernel ratio: $\rho_i \leq \rho_i \leq \bar{\rho}_i$. Exact zeros known from rank $K_\gamma \leq W + 1$ may have both endpoints set to zero. No target or optimization trajectory enters these spectral bounds.

A.1 DERIVATION OF THE EXPLICIT GAMMA DEPENDENCE

The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ gives

$$\int_{\mathbb{R}} [1 - \tanh(\gamma(z_a - c)) \tanh(\gamma(z_b - c))] dc = 2d_{ab} \coth(\gamma d_{ab}). \quad (12)$$

Multiplication by $Q^\top$ and Q removes the constant term and proves (4). The integral is finite after projection because $Q^\top t_\gamma(c)$ decays at both ends.

The smoothing interpretation can also be made exact. With $s_\gamma(x) = \frac{\gamma}{2} \operatorname{sech}^2(\gamma x)$, $\tanh(\gamma \cdot) = s_\gamma * \operatorname{sign}$. Its Fourier multiplier is

$$A_\gamma(\omega) = \widehat{s}_\gamma(\omega) = \frac{\pi|\omega|/(2\gamma)}{\sinh(\pi|\omega|/(2\gamma))}, \quad A_\gamma(0) = 1. \quad (13)$$

For completeness, substitution $r = e^{2\gamma x}$ in the Fourier transform gives $\int_0^\infty r^{-i\omega/(2\gamma)}(1+r)^{-2} dr$. The beta integral evaluates this as $\Gamma(1 - i\omega/(2\gamma))\Gamma(1 + i\omega/(2\gamma))$, which equals (13) by the reflection formula. For $u = Qv$, the step combination $\sum_a u_a \operatorname{sign}(c - z_a)$ has compact support, and its derivative is $2 \sum_a u_a \delta_{z_a}$. Parseval gives

$$v^\top H_\gamma v = \frac{2}{\pi h M} \int_{\mathbb{R}} \frac{A_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega z_a} \right|^2 d\omega. \quad (14)$$

The apparent zero-frequency singularity is removable since $\sum_a (Qv)_a = 0$. Thus gamma acts on a direction's frequency content through the explicit weight A_γ^2 . Since $t/\sinh t$ decreases for $t > 0$, increasing gamma increases every directional response of H_γ and hence its ordered eigenvalues. This order concerns the integral kernel; exterior subtraction and normalization need not preserve monotonicity of every finite ratio.

A.2 TRANSFER TO THE FINITE DICTIONARY

For $u \perp \mathbf{1}$, the function $g(\zeta) = \sum_a u_a \tanh(\gamma(\zeta - z_a))$ is analytic for $|\operatorname{Im} \zeta| < \pi/(2\gamma)$. Subtracting the common translate at $\bar{z}$ and integrating derivatives, Minkowski and Cauchy-Schwarz yield

$$\int_{\mathbb{R}} |g(t \pm i\theta/\gamma)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \theta \|z - \bar{z}\mathbf{1}\|^2 \|u\|^2. \quad (15)$$

Here $|\operatorname{sech}^2(t + i\theta)| \leq \sec^2 \theta \operatorname{sech}^2 t$ and $\int \operatorname{sech}^4 t dt = 4/3$. Shift the contour for the analytic function $g(\zeta)^2$, not $|g(\zeta)|^2$: its transform is bounded by the right side of (15) times $e^{-\theta|\omega|/\gamma}$. Poisson summation at frequencies $2\pi k/h$ then gives

$$\left\| \frac{1}{M} \sum_{c \in c_0 + h\mathbb{Z}} Q^\top t_\gamma(c) t_\gamma(c)^\top Q - H_\gamma \right\| \leq \delta_\gamma. \quad (16)$$

The geometric sum over $k \neq 0$ produces the denominator in (6).

Deleting exterior centers converts the infinite lattice to the actual dictionary. The retained deletions give T_γ . For a remaining center $c \geq c_R$, projection removes the constant limit and $1 - \tanh t \leq 2e^{-2t}$ gives an outer-product norm, divided by M , of at most $4e^{-4\gamma(c-z_{\max})}$. Summing this geometric series and its left-tail counterpart bounds the remaining positive deletion by $\tau_\gamma I$. This proves (8).

If κ_i are the eigenvalues of $Q^\top K_\gamma Q$, min–max and codimension-one interlacing give

$$\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma, \quad \mu_i \geq \kappa_i \geq \mu_{i+1}. \quad (17)$$

The Rayleigh quotient gives $\mu_1 \geq \ell_\gamma$. In the decomposition $\text{span}(v_0) \oplus v_0^\perp$, the diagonal blocks of K_γ are bounded above by ℓ_γ and $c_\gamma I$, and the off-diagonal block has norm b_γ . Its quadratic form is therefore bounded by the largest eigenvalue of $\begin{bmatrix} \ell_\gamma & b_\gamma \\ b_\gamma & c_\gamma \end{bmatrix}$, namely L_γ . Dividing the interlacing bounds by these normalization endpoints proves the relative-rate enclosure from (11).

A.3 CLASSICAL OUTPUT-ERROR CONSEQUENCE

For $r_n = \Phi_\gamma w_n - y$, normalized least-squares GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$ and $r_0 = -y$. Expansion in the actual finite-kernel eigenvectors gives

$$e_n^2 = \sum_i p_i (1 - \eta \mu_i)^{2n} = \sum_i p_i (1 - \rho_i/2)^{2n} \geq \sum_i p_i (1 - \bar{\rho}_i/2)^{2n}. \quad (18)$$

The last inequality follows because the factor decreases with $\rho_i \in [0, 1]$. This proves Theorem 3.2. Our step is a conservative fraction of the stability limit $2/\mu_1$; any nonoscillatory step $0 < \eta \mu_1 \leq 1$ instead uses $\eta \mu_1 \rho_i$. Only exact nullspace energy gives a permanent floor. For $\mu_i > 0$, $u_i = \Phi_\gamma(\Phi_\gamma^\top u_i / (M \mu_i))$ is representable, however small its rate. The target weights require the actual eigenvectors; the theorem bounds eigenvalues, not their target alignment.

B Evaluation, accuracy, and empirical scope

Which quantities are computed. The theorem endpoints use the explicit center integral, exterior-center subtraction, and scalar normalization bounds. A separate SVD of the actual finite feature matrix supplies the validation eigenvalues and target weights p_i . The plotted lower error uses the theorem’s upper rates, not actual rates. The spectral figure selects three ranks for readability; the error bound includes every numerically resolved target projection.

The integral is evaluated through a positive quadrature factor rather than by subtracting nearly equal entries of (4). There is no Fourier truncation. Truncating the center integral at $\pm(\max_a |z_a| + P/\gamma)$ omits a positive matrix of norm at most $2e^{-4P}/(h\gamma)$, added to the numerical upper endpoint. The exterior remainder enters the lower endpoint. Compressing the combined integral and exterior factors Z to rank r introduces the two-sided bound $2\sigma_1(Z)\sigma_{r+1}(Z) + \sigma_{r+1}(Z)^2$ on their signed Gram matrix. Quadrature refinement and an empirical arithmetic guard are checked separately; the figures are FP64 evaluations, not interval-arithmetic certificates. Refining quadrature order from 10 to 16 and padding from 20 to 24 changes the lower error curves by at most 9.6×10^{-11} .

The remaining analytic slack comes from the interlacing index shift, the normalization bounds ℓ_γ, L_γ , and additive allowances. Small absolute errors can dominate extremely small eigenvalues; the enclosure is valid but need not be informative for every geometry or mode. In this experiment it is informative both for the plotted ratios and for the full error trajectory. Omitting unresolved target energy from the lower curve only weakens it; it is not interpreted as an exact approximation floor.

Target, geometry, and measurements. All displayed data use

$$f(x) = \frac{\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(14\pi x)}{\sqrt{21/32}}, \quad x \in [-1, 1]. \quad (19)$$

The total hidden width is $W = 512$, comprising $N = 467$ interior cells and $R = 22$ halo centers on each side, so $h = 2/467$ and $W = N + 2R + 1$. There are 2,048 uniform training midpoints, 4,096 validation points, and 8,192 dense evaluation points. Every reported error is relative output L^2 error on its stated grid, not squared loss or an EMA. The dense grid checks resolution; it is not an untouched generalization test. Frozen GD starts its readout at zero and executes five million full-batch FP64 updates with $\eta = 1/(2\mu_1)$ at each bandwidth. Figure 4A displays training error. Its spectral reference agrees with executed GD within 2.9×10^{-14} .

Joint training optimizes all slopes, centers (through affine biases), and readout parameters for five million updates on five paired seeds. One recipe per optimizer is selected by median final validation error over all five seeds. The selected Adam and GD schedules are full-horizon cosine decays with initial learning rates 0.05 and 0.2, respectively. Frozen Adam uses the same final-validation selection criterion. Panel B shows raw training errors summarized for display; quoted joint and frozen-Adam endpoint values use the dense grid. Display bins retain extrema, including spikes, rather than replacing the comparison with first tolerance crossings. Panel C measures $h|a_j|$ for learned features $\tanh(a_j x + b_j)$, using the original reference spacing h .

What the joint experiment establishes. Adam’s median scaled RMS slope is 0.0582, and its 99th percentile is 0.2466; the five seeds have 6, 5, 5, 4, 7 features at or above $1/4$. These statistics describe heterogeneous geometry, not an effective uniform bandwidth. The uniform $1/4$ dictionary is an empirical comparison, not a necessary scale for every target or learned dictionary.

Joint Adam continues improving late in training; its median endpoint error falls about 28% between the two- and five-million-update experiments. The result is a precision gap under a large finite budget, not permanent failure. Checked direct fits and an additional two-million-update readout restart on the learned five-million-update features barely improve their error. Multiplying the learned slopes by 4 or 16 at fixed centers worsens the median refitted error to 0.0916 or 0.139, respectively. These interventions change nonnested feature spaces: they support investigating slopes and centers together, not a theorem that increasing slopes always helps. Neither the fixed-kernel proof nor its scalar GD rates apply to evolving features or Adam’s adaptive preconditioner. The next theoretical question is how evolving features acquire a useful geometry.

Reproducibility. The source and numerical provenance for the new spectral figure are `experiments/expD36_frozen_gamma_probe/section34_spectrum_figure.py` and `output/diagnostics/section34_long_horizon/main_5m/spectrum/relative_rates.json`. The existing three-panel figure, full protocols, checks, and evidence links are indexed in `output/diagnostics/section34_long_horizon/README.md`. Both figures reuse completed runs; preparing this note launches no new training experiments.

References

- Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://papers.neurips.cc/paper_files/paper/2020/hash/55053683268957697aa39fba6f231c68-Abstract.html.
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